



Mae Fah Luang University

School of Applied Digital Technology



Computer Engineering

TOF.2 | TOF.3 | TABEE 2567



Bachelor of Engineering Program in Computer Engineering

Degree Title:

- **Full Title:** Bachelor of Engineering (Computer Engineering)
- **Abbreviation:** B.Eng. (Computer Engineering)



Program Overview: The Bachelor of Engineering Program in Computer Engineering aims to produce graduates equipped with world-class theoretical knowledge and practical expertise. Our students develop specialized skills in designing, planning, and applying emerging technologies to create innovative solutions. The curriculum covers the comprehensive maintenance of both computer hardware and software systems.

To meet the demands of modern industries and the private sector, the program offers **three specialized learning tracks** tailored to students' interests:

1. **Hardware and Intelligent Control Systems**
2. **Data Science and Artificial Intelligence (AI)**
3. **Data Communications and Networks**

Our graduates are trained to integrate computer systems into industrial applications, possess strong self-learning capabilities for lifelong education, and are prepared to create engineering masterpieces as future technology leaders.



Career Opportunities

- Computer Engineer



- Electronics Control Engineer
- Data and AI Engineer
- Software Developer
- Network Engineer
- Telecommunications Engineer
- Innovation Entrepreneur



เชี่ยวชาญด้านอินเทอร์เน็ตของสรรพสิ่งและเมืองอัจฉริยะ

Further Study Opportunities

- Graduates are well-prepared to pursue Master's and Doctoral degrees both in Thailand and at leading international institutions. Opportunities for advanced studies include Computer Engineering or other closely related fields, such as: Electrical Engineering, Electronics Engineering, Telecommunications Engineering, And other relevant technological disciplines.



“Why Computer Engineering at MFU?”

- **English-Medium Instruction:** Enhancing global career prospects and ensuring readiness for the international job market.
- **Industry-Aligned Curriculum:** Regularly updated courses specifically designed to meet the evolving demands of modern employers.
- **State-of-the-Art Laboratories:** Access to fully equipped, high-tech facilities that support advanced learning and experimentation.
- **Entrepreneurial Programming Excellence:** Cultivating professional-level coding expertise, enabling students to develop their own software and innovative products.
- **Problem-Based Learning (PBL):** A hands-on approach that bridges the gap between theory and practice, making learning both engaging and expert-driven through real-world challenges.
- **Work-Integrated Learning:** Gain professional experience and earn income through industry placements during your studies.
- **Elite Strategic Partnerships:** Strong collaborations with renowned public and private IT organizations and global tech giants.



Program Slogan

- "MFU Engineering: Hardworking, Hands-on, Heart-centered, Highly Skilled."

Learning Path

- **Comprehensive Theory and Practice:** Our curriculum integrates foundational theory with hands-on expertise in computer engineering, including **Programming, Embedded Systems Design, Computer Networks & Security**, and **Artificial Intelligence (AI)**.
- **Real-World Project Integration:** Students are empowered to tackle complex projects that emphasize the practical application of knowledge in real-world scenarios.
- **Specialized Career Tracks:** Students can tailor their professional development through three specialized tracks:
 - Hardware and Embedded Systems
 - Data Communications and Networking
 - Data Science and Artificial Intelligence



Unique Learning Experience

- **Cutting-Edge Technology & Tools:** Mastery of modern technologies, including **Embedded Systems, IoT, Network Security, Cloud Computing, AI, and Machine Learning.**
- **Innovative Ecosystem:** A vibrant learning environment designed to foster creativity and innovation, fully supported by a faculty of industry experts.

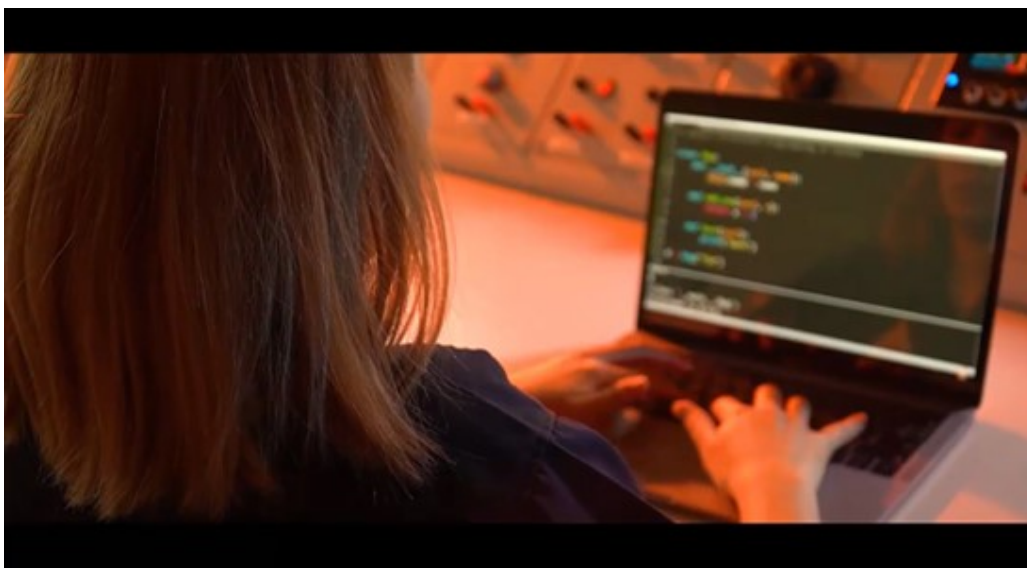
Key Highlights of the Program

- **Outcome-Based Education (OBE):** A curriculum strategically designed to prioritize student competencies and meet the evolving demands of the global labor market.
- **Focus on Emerging Technologies:** Intensive emphasis on applying the latest advancements in both software and hardware.
- **Global Internship Opportunities:** Access to prestigious internships with leading technology firms, both domestically and internationally.
- **International Academic Exchange:** Opportunities to broaden perspectives through exchange programs at top-tier universities worldwide.



Is This the Right Fit for You?

- **The Innovator:** For those who thrive on problem-solving and creating groundbreaking innovations through technology.
- **The Tech Enthusiast:** For those passionate about programming and system design—ranging from Embedded Systems and Network Security to IoT, Data Science, and AI.
- **The Future Leader:** For those aspiring to lead in fast-growing industries, specifically within the IT, AI, and IoT sectors.
- **The High-Demand Professional:** For those seeking careers in high-impact fields such as Network Security, Embedded Systems, and Data Science.



Collaboration with Leading Industries



HUAWEI TECHNOLOGIES
(THAILAND)



Palo Alto Networks
Thailand



Cisco Systems
(Thailand)

Student Activities and Successes



The Winner of the network track in Huawei ICT Competition Thailand
2024-2025!

Team: Union won 1st prize/Thailand's representative in the Asia-
Pacific regional competition:

- Min Thway Khant
- Hein Htet Aung
- Kanyakrit Bowonsuwan

Grand Prize: 100,000 THB & Huawei Matepad 11.5 inch + an
opportunity to represent Thailand at the Regional Round in Kuala
Lumpur, Malaysia

Team Noobs secured 2nd prize:

- Myo Thet Paing



- Jayvert Dale Ancheta

- Zaw Linn Lhyo

1st Runner-Up: 40,000 THB & Huawei Freebud 6i Skullpanda edition



Ms. Piriya Korn Chonpattanapaian – 4th-year Computer Engineering student

full scholarships to participate in the ASEAN Cyber Shield Exchange Student Programme 2025 at Kangwon National University in the Republic of Korea from February to December 2025

Ms. May Htet Htet Khine, a 4th-year Computer Engineering

student, presented her research paper, "Particle Weight

Redistribution in Particle Filtering to Improve Nonlinear



State Estimation," at The 8th International Conference on
Information Technology (InCIT 2024), held on November
14–15, 2024, in Chonburi, Thailand.



Mr. Simeon Thosawongwat, a third-year Computer Engineering student, in collaboration with Ms. Kanyanat Somthong, a third-year Environmental Health student from the School of Health Science, received First Runner-Up in the 24th Thailand Innovation Awards for their project: "Demeter: A Precision Agriculture Unmanned Aerial Vehicle."

They were awarded a trophy of honor, a certificate of recognition, and a cash prize of 30,000 baht.



Program Philosophy

- Computer Engineering has played a pivotal role in industrial transformation and socio-economic development from the past to the present. It remains a vital discipline for navigating the disruptions of today and the future. The continuous invention of new theories and engineering innovations is an urgent necessity, achieved through the strategic development of human resources in this field.

Our Educational Foundation: Constructivism This curriculum is built upon **Constructivist Theory** as its core foundation, integrating both **Cognitive Constructivism** and **Social Constructivism**. We believe that learning is an internal process where students actively construct their own knowledge. By connecting new information and environmental experiences with their existing understanding, students develop their own **Cognitive Structures**.

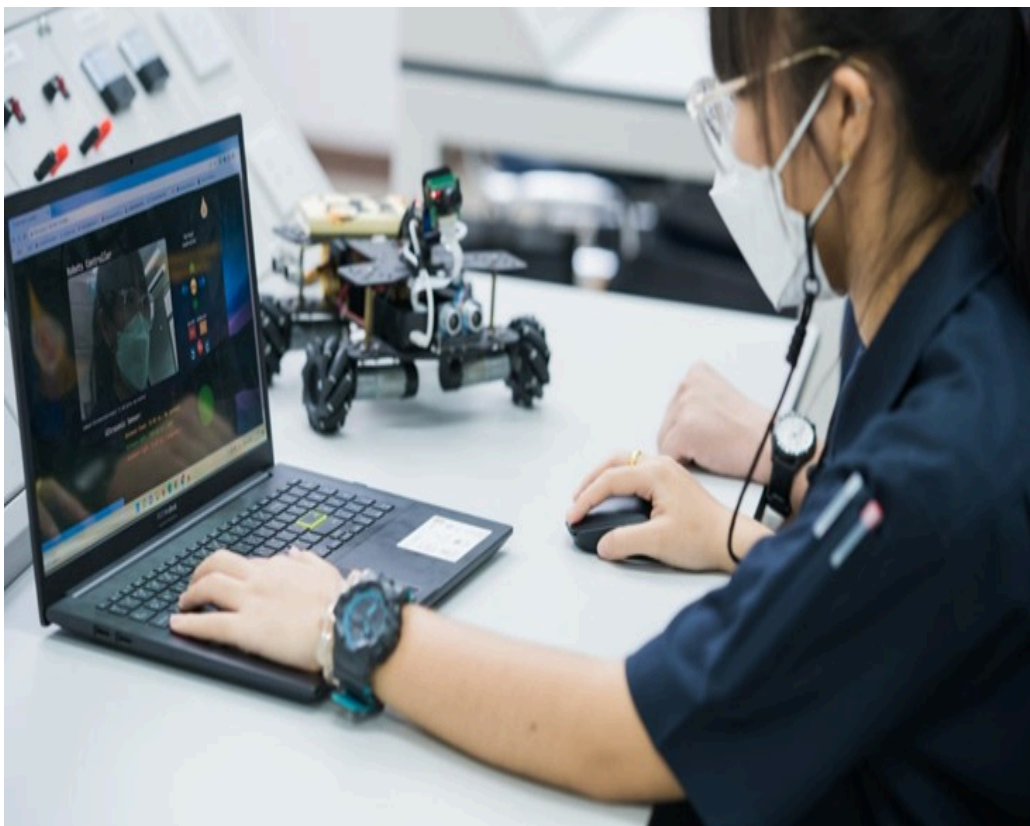
To stimulate this growth, we utilize problem-based scenarios and social interactions to create "cognitive conflict." This challenges students to engage in deep reflection and problem-solving, ultimately leading to the formation of advanced cognitive frameworks. Rather than traditional knowledge transmission, our teaching focuses on **knowledge construction**.

Pedagogical Approach: Problem-Based Learning (PBL) The program primarily employs **Problem-Based Learning (PBL)**. We place students in real-world situations, encouraging them to discover solutions independently through the guidance of facilitators and collaborative interaction with peers.



Our Mission The Bachelor of Engineering Program in Computer Engineering at Mae Fah Luang University is committed to producing graduates with world-class theoretical and practical expertise. Our graduates are lifelong learners with versatile skills in design, research, and system development. By leveraging **Emerging Technologies** across three key pillars:

- **Hardware and Embedded Systems**
- **Data Science and Artificial Intelligence (AI)**
- **Data Communications and Networks**
- They are prepared to solve complex engineering challenges and meet the evolving demands of modern industry and the private sector.



Program Significance

- The Bachelor of Engineering Program in Computer Engineering at Mae Fah Luang University has been developed in strict accordance with the **Higher Education Act of 2019** and Thailand's **20-Year National Strategy (2017–2036)**. As digital transformation directly impacts the nation's economy, society, and culture, developing human capital to meet these evolving demands is an urgent and vital mission.



Driving National Competitiveness This program is dedicated to producing high-caliber scholars and computer engineers capable of conducting advanced research and developing innovations. By enhancing the nation's competitive potential amidst global disruptions, our graduates serve as the essential workforce driving the country's future.

Outcome-Based Education (OBE) We employ an **Outcome-Based Education (OBE)** approach to ensure that students not only possess strong professional ethics but also achieve profound mastery in Computer Engineering. The curriculum places a strategic emphasis on **Emerging Technologies** across three specialized pillars:

- **Hardware and Embedded Systems**
- **Data Science and Artificial Intelligence (AI)**
- **Data Communications and Networks**
- **Leadership in a Globalized Society** Beyond technical excellence, the program fosters a positive mindset and adaptability. Our students are prepared to thrive and collaborate within a **multicultural society**, empowering them to become resilient leaders capable of navigating any global disruption.



Program Educational Objectives (PEOs)



The Bachelor of Engineering Program in Computer Engineering aims to produce graduates who achieve the following objectives:

- **PEO 1: Effective Communication and Collaborative Leadership** To produce graduates who possess a profound understanding of computer engineering principles, demonstrate high-level communication skills, and effectively collaborate within diverse teams as both responsible leaders and proactive team members.
- **PEO 2: Lifelong Learning, Ethics, and Adaptability** To produce graduates who utilize their knowledge and skills for continuous self-development and lifelong learning. Our graduates are expected to adapt resiliently to global shifts and societal changes while upholding high moral standards, professional ethics, and integrity.
- **PEO 3: Technical Mastery in Engineering Design and Development** To produce graduates with advanced expertise in designing and developing comprehensive computer engineering systems, spanning across **Hardware, Software, Data Science, and Network Systems**.



Program Learning Outcomes (PLOs)

- **PLO1:** Demonstrate ethical behavior and moral integrity, respecting organizational and social regulations while adhering strictly to computer ethics and legal standards.



- **PLO2:** Articulate and integrate principles, theories, and foundational knowledge in computer engineering and emerging technologies to effectively resolve engineering challenges.
- **PLO3:** Design computer engineering systems by applying foundational engineering knowledge, emerging technologies, and relevant interdisciplinary sciences.
- **PLO4:** Demonstrate the ability to work effectively in teams within a multicultural society.
- **PLO5:** Utilize numerical analysis skills, effective communication, and strategic selection of emerging technologies to solve computer engineering problems appropriately.
- **PLO6:** Develop efficient systems to solve computer engineering problems derived from real-world industrial challenges.

Curriculum Structure

Total Credits Required: 132 Credits (4-Year Program)	
1. General Education Courses	30 Credits
2. Specialized Courses	87 Credits
2.1 Professional Foundation Courses	30 Credits
2.2 Major Required Courses	36 Credits
2.3 Major Elective Courses	21 Credits
3. Free Elective Courses	6 Credits
4. Field Experience / Internship	9 Credits

Study Plan

ความเชื่อมโยงระหว่างรายวิชาและแผนการศึกษา

= วิชาศึกษาทั่วไป
 = วิชาพื้นฐานวิชาชีพ
 = วิชาชีพบังคับ
 → = รายวิชาบังคับก่อน
 = วิชาชีพเลือก
 = วิชาเลือกเสรี
 = วิชาประสบการณ์ภาคสนาม
 --> = แนะนำให้เรียนมาก่อน

รายวิชาเรียน หลักสูตรวิศวกรรมศาสตรบัณฑิต สาขาวิชาวิศวกรรมคอมพิวเตอร์							
ชั้นปีที่ 1		ชั้นปีที่ 2		ชั้นปีที่ 3		ชั้นปีที่ 4	
ภาคเรียน 1	ภาคเรียน 2	ภาคเรียน 3	ภาคเรียน 4	ภาคเรียน 5	ภาคเรียน 6	ภาคเรียน 7	ภาคเรียน 8
1005120 Thai for Com	1006135 Eng Com 2	1501208 Database	1501207 Math for Eng 1	2408101 Chinese 1	2408102 Chinese 2	1601114 Global Citizen	1501496 Co-op Edu บวช
1006134 Eng Com 1	1001127 Psychology	1501206 Math for Eng 2	1501209 Com Arch	1501315 Software Proj	1203102 MO and Enterp	1501402 CE Project	1501497 Overseas train บวช
1301110 Intro to IT	1501114 Math for Eng 1	1501211 Fund Digital	1501221 Embedded Sys	1305216 Mobile App	1501401 CE Pre-Project	1501xxx Major Elective 6	1501498 Research Proj
1101105 Sci and Math	1501118 Data Structure	1501410 Intro Data Sci	1501219 AI Application	1501xxx Major Elective 1	1501xxx Major Elective 4	1501xxx Major Elective 7	
MFU333 MFU Char	1501117 OOP	1501212 Data Comm	1501220 IT and Cyber	1501xxx Major Elective 2	1501xxx Major Elective 5	1xxxxxx Free Elective 2	
1501115 Com Eng Ics	1501108 Electric Circuit	1501112 Eng Program	1501312 Web App	1501xxx Major Elective 3	1xxxxxx Free Elective 3		
1501116 Com Program							

Major Elective Groups

Engineering Program in Computer Engineering

วิศวกรรมศาสตรบัณฑิต สาขาวิชาวิศวกรรมคอมพิวเตอร์

ฮาร์ดแวร์และระบบ
ควบคุมอัจฉริยะ
Hardware and
Intelligent Control
Systems

Data Science and
Artificial
Intelligence
วิทยาการข้อมูลและ
ปัญญาประดิษฐ์

Data
Communications
and Networks
เครือข่ายและ
การสื่อสารข้อมูล



Level	Track Course		
	Data Science and Artificial Intelligence	Data Communications and Networks	Hardware and Embedded Systems
Beginner (Required) 	• Introduction to Data Science • Artificial Intelligence Applications	• Data Communications and Networks • Information Technology and Cybersecurity	• Fundamentals of Digital Logic and Microcontrollers • Embedded System and Internet of Things
Moderate (ME) 	• Machine Learning • Data Engineering • Statistical Analysis for Data Science • Computer Vision and Visual Analytics	• Service Provider Network • Network Programming and Automation • Cloud Computing • Electronic Communication Systems • Enterprise Wireless Technologies and Infrastructure. • Network System Security • IP Network Design	• Fundamental of Electrical Engineering • Model Based Design • Intelligent Control System • Signal Processing for Intelligent tracking System • Electronics and Electric Drives
Advance (ME) 	• Social Network Analysis • Natural Language Engineering • Business Data Analytics • Big Data Management and Analytics • Automatic Speech Recognition	• Web Application Security • Communication and Computer Engineering Applications • UNIX/Linux for Enterprise Environment • Enterprise Networks Infrastructure • Data Center Infrastructure • Network Analytics and Telemetry	• Computer Control Systems • Biomedical Signal Processing and Control • Robotics and Applications • Sequence Control and PLC • Optimization Technique

Tuition & Fees

- Total Program Fee: 224,000 THB
- Tuition Fee per Semester: 28,000 THB

[View Full Course Details - Click Here!](#)

References & Updates

- Based on: Bachelor of Engineering Program in Computer Engineering (2021 Curriculum)

Last Updated: June 17, 2025



Bachelor's Degree

Digital and Communication Engineering

Computer Engineering

Digital Technology for Business Innovation

Software Engineering

Multimedia Technology and Animation



School of Applied Digital Technology Mae Fah Luang University

Contact Us

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ระเบิดไอดีสุดล้ำ! 🚀 ประมวลภาพบรรยากาศ
ความสนุกจากงาน MediHack Junior 2026 🏆

สำนักวิชาเทคโนโลยีดิจิทัลประยุกต์ (Applied Digital
Technology) ได้จัดกิจกรรม MediHack Junior
2026 ขึ้น ณ อาคารเรียน 11 เมื่อวันที่ 18 เมษายน 2569